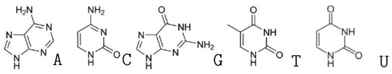
BMEG3102 - Bioinformatics

Bioinformatics: Application of computer science and information technology to the field of biology and medicine. Suitable for computational problems with: (i) large data size, (ii) difficult computational problems.
Cell: Basic functional unit of life.

Chromosome
In human, each DNA-containing cell has 23 pairs of chromosomes (one from father, one from mother).
Mitosis: each chromosome is duplicated and both daughter cells have the complete set of chromosomes.
Meiosis: each germ cell contains only one of each pair of chromosomes

Two Copies of Each Chromosome
More combinations: For each of the 23 pairs of chromosomes, only one is passed to each offspring, which creates 223 possible combinations.
Error tolerance: If one copy has problem, there is still another copy.
Evolution: Having one normal copy, the other is more free to change, sometimes resulting in an overall advantage.

DNA (DeoxyriboNucleic Acid)
Nucleotides: joined together through strong phosphate to form two strands
Pyrimidines: Cytosine (C) and Thymine (T) have 1 ring
Purines: Adenine (A) and Guanine (G) have 2 rings



Each nucleotide unit has three components:
Nitrogenous base, Pentose sugar (ribose), Phosphate
Genes: Regions with any heritable trait
Transcription: DNA transcribes to RNA (Ribonucleic)
DNA serves as template, "introns" are spliced and leaving only "exons" in mRNA.
Translation: RNA translates to protein
mRNA enters the ribosome; amino acids are delivered by tRNA. UTRs (untranslated regions) determine the start and end of translation. Every three nucleotides form a "codon" unit.

Sequence Similarity
Evolutionarily: Either diverged from a common ancestor for a shorter time, or preservation of original properties
"Selective pressure" – conservation suggests importance (changes are unfavorable)
Molecular level: Structures are more similar; Containing similar functional units (domains)

Sequence Alignment
Given a set of sequences, an alignment is the same set of sequences with zero or more gaps inserted into them so that 1. They all have the same length afterwards; 2. For each column (also called a position or a site), at least one of the resulting sequences is not a gap.
Number of possible alignments: Given two sequences with m,n ($m \le n$). Then, total number of possible alignments is

$$\sum_{x=n-m}^{n}\binom{x+m}{x}\binom{m}{x+m-n}$$

Needleman-Wunsch Global Alignment Algorithm
Let $r[m+1]=\emptyset, s[n+1]=\emptyset$. Define $V(i,j) =$ optimal alignment score of suffixes $r[i \dots m], s[j \dots n]$. Optimal alignment score: $V(1,1)$.
Scoring scheme: match $+M$, indel $-I$, mismatch $-P$

$$V(i,j)=\max\begin{cases} V(i+1,j+1)+M, \text{if } r[i]=s[j] \\ V(i+1,j+1)-P, \text{if } r[i]\neq s[j] \\ V(i,j+1)-I, \text{(adds an indel to } r[i]) \\ V(i+1,j)-I, \text{(adds an indel to } s[j]) \end{cases}$$

Add an arrow from $V(i',j')$ to $V(i,j)$ that maximizes $V(i,j)$.
Improvement: Use a scoring matrix.

Affine gap penalty
$y=-a-bx$, where x is the size of gap
$V(i,j)=\max\{V_0(i,j),V_1(i,j),V_2(i,j)\}$
$V_0(i,j)=V(i+1,j+1)+\sigma(r[i],s[j])$

$$V_1(i,j)=\max\begin{cases} V_0(i+1,j)-a-b \\ V_1(i+1,j)-b \\ V_2(i+1,j)-a-b \end{cases}$$

$$V_2(i,j)=\max\begin{cases} V_0(i,j+1)-a-b \\ V_1(i,j+1)-a-b \\ V_2(i,j+1)-b \end{cases}$$

$V(m+1,n+1)=V_1(m+1,n+1)=V_2(m+1,n+1)=0$, $V_0(k_1,n+1)=V_0(m+1,k_2)=V_1(m+1,j)=V_2(i,n+1)=-\infty, V_1(i,n+1)=-a-(m-i+1)b, V_2(m+1,j)=-a-(n-j+1)b$, for $k_1\in[1,m+1], k_2\in[1,n+1], i\in[1,m], j\in[1,n]$.
Dynamic Programming: (i) Divide-and-Conquer; (ii) Repeated subproblems.

Smith-Waterman Local Alignment Algorithm
Local alignment is a set of *subsequences* each extracted from one of the original sequences, with zero or more gaps inserted into these subsequences so that 1. They all have the same length afterwards 2. For each position, at least one of the extracted sub-sequences is not a gap.
Define $V(i,j) =$ optimal alignment score of among all prefixes of the suffixes $r[i \dots m], s[j \dots n]$.

$$V(i,j)=\max\begin{cases} V(i+1,j+1)+\sigma(r[i],s[j]) \\ V(i,j+1)+\sigma('\_'+s[j]) \\ V(i+1,j)+\sigma(r[i],'\_') \\ 0 \end{cases}$$

Optimal score: $\max\{V(i,j)\}$. Output the alignment when encountering a score of 0.

Heuristics Methods
Short subsequences with high similarity; Combines and refine initial results.
Dot plot: gives information about conserved regions, non-conserved regions, inversions, insertions and deletions, local repeats, multiple matches, translocations.
Stretched *diagonals* marks a local *exact* match.
Limitations: must be exact matches, space consumption, resolutions, not quantitative.

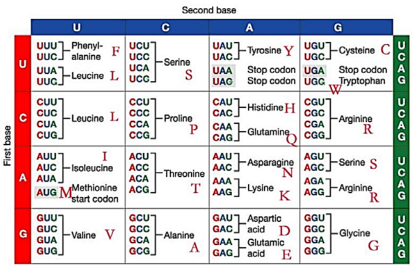
FASTA
Step 1: Break the search sequence into words with size k.
Step 2: Construct a table for *query* sequence showing the *locations of each word* within the sequence.
Step 3: Construct the second table that compares each amino acid in the *target* sequence with the query sequence. It records the offset of the target sequence.
Step 4: Areas of identity can be found quickly by large instances of the same offset. Combine subalignments, rescore and select the best alignments based on the substitution matrix.
Step 5: Perform local alignment in narrowed regions. FASTA miss an optimal alignment when k is too large, high penalty for mismatches, or too many local candidates.

FASTA File Format: ID + Less than 80 characters per line

```
>SEQUENCE_1
MTEITAAMVKELRESTGAGMMDCKNALS
>SEQUENCE_2
SATVSEINSETDFVAKNDQFIALTKDTT
```

BLAST (Basic Local Alignment Search Tool)
Allows for high-scoring inexact matches, extends stretches, evaluates statistical significance (E-value, expected number of matches with score Q or larger for a pair of random sequences of length m and n, smaller=unlikely).
Step 1: breaks query sequence into words (4 by default), words composed mostly of common amino acids will be discarded.
Step 2: search for word occurrences in the database, extend the match in both directions until the local alignment score falls below a fixed threshold.

Variations Depending on the Query-Database
Nucleotide-nucleotide BLAST (blastn); Protein-protein BLAST (blastp); Nucleotide 6-frame translation-protein BLAST (blastx, perform 6-frame translation of nucleotide query, then compare with protein sequences in database); Protein-nucleotide 6-frame translation BLAST (tblastn, compare query protein sequence with 6-frame translation of nucleotide sequences in database); Nucleotide 6-frame translation-nucleotide 6-frame translation BLAST (tblastx, perform 6-frame translation of query and database nucleotide sequences, then perform comparisons)



Iterative Database Search
Use BLAST to obtain highly similar sequences and then construct a profile of the sequences. Apply the profile to BLAST. Repeat until no more sequences are returned.

Multiple Sequence Alignment
Alignment score measure: (i) all pairs average score, (ii) consensus sequence averaging, PWM.
Technique: Clustal, T-Coffee, MAFFT, MUSCLE
Clustal: distance matrix between all pairs: $\text{dist} = \text{length of alignment} - \text{alignment score}$, tree construction: perform hierarchical clustering and progressively align the sequences based on the tree.

Evolutionary Distance: Number of mutations happened since divergence. Assumption: sites dependent, mutation rates are constant for different sites and over time.
Estimating αt by p_{diff} is enough, empirically x/n.

Jukes-Cantor Mutation Model
Equal rate of substitution, α, to other three bases in one unit of time. Define $\mathbb{P}_{X\to Y}(t)$ as the probability that a base was X at time 0, and Y at time t.

$$\mathbb{P}_{X\to X}(t)=\frac{1}{4}+\frac{3}{4}e^{-4\alpha t}, \mathbb{P}_{X\to Y}(t)=\frac{1}{4}-\frac{1}{4}e^{-4\alpha t}.$$

Let K_{sup} be the no. of substitutions *per site* happened to the two sequences since their divergence.

$$\mathbb{E}[K_{\text{sup}}]=3\alpha t=-\frac{3}{4}\ln\left(1-\frac{4x}{3n}\right),$$

$$\text{Var}[K_{\text{sup}}]=\frac{p_{\text{diff}}-(p_{\text{diff}})^2}{n(1-\frac{4}{3}p_{\text{diff}})}=\frac{x/n-(x/n)^2}{n\left(1-\frac{4x}{3n}\right)^2}$$

Kimura Two-Parameter Mutation Model
Transitions are more frequent than transversions ($\alpha > \beta$).

$$\mathbb{P}_{X\to X}(t)=\frac{1}{4}+\frac{1}{4}e^{-4\beta t}+\frac{1}{2}e^{-2(\alpha+\beta)t}, \mathbb{P}_{A\to C}(t)=\mathbb{P}_{A\to T}(t)=\dots=\frac{1}{4}-\frac{1}{4}e^{-4\beta t},$$

$$\mathbb{P}_{A\to G}(t)=\mathbb{P}_{G\to A}(t)=\mathbb{P}_{C\to T}(t)=\mathbb{P}_{T\to C}(t)=\tfrac{1}{4}+\tfrac{1}{4}e^{-4\beta t}-\tfrac{1}{2}e^{-2(\alpha+\beta)t}.$$

Let $p_{\text{diff1}}=x_1/n, p_{\text{diff2}}=x_2/n$ (observed transitions, observed transversions)

$$\mathbb{E}[K_{\text{sup}}]=-\frac{1}{2}\ln\left(\frac{1}{1-2p_{\text{diff2}}-p_{\text{diff2}}}\right)+\frac{1}{4}\ln\left(\frac{1}{1-2p_{\text{diff2}}}\right)$$

$$\text{Var}[K_{\text{sup}}]=\frac{1}{n}\left[p_{\text{diff1}}\left(\frac{1}{1-2p_{\text{diff1}}-p_{\text{diff2}}}\right)^2+p_{\text{diff2}}\left(\frac{1}{2-4p_{\text{diff1}}-2p_{\text{diff2}}}+\frac{1}{2-4p_{\text{diff2}}}\right)^2 -\left(\frac{p_{\text{diff1}}}{1-2p_{\text{diff1}}-p_{\text{diff2}}}+\frac{p_{\text{diff2}}}{2-4p_{\text{diff1}}-2p_{\text{diff2}}}+\frac{p_{\text{diff2}}}{2-4p_{\text{diff2}}}\right)^2\right]$$

Substitution Matrix for Amino Acid Substitutions
Point Accepted Mutation Model (PAMx): each element records the *probability* of that substitution given a *mutation rate* of *x substitution per 100 amino acids*.
Similar chemical properties means higher likelihood of substitution.
$\mathrm{V}\to\mathrm{L}$ more likely than $\mathrm{V}\to\mathrm{I}$: (i) path exists, (ii) I is a source, L is a sink.
BLOSUM: Blocks of amino acid substitution matrix, based on local alignments of *very conserved regions*. The BLOSUMy matrix, the *local alignments involve sequences that are more than y% identical*.
Larger $y \Rightarrow$ less substitutions. Each entry contains the *log-odd* score: $S_{ij}=(1/\lambda)\log p_{ij}/(p_i p_j)$, where p_{ij} is the fraction of observed substitutions between amino acids i and j, p_i and p_j are the fraction of sites with amino acids i and j.

Phylogeny
A systematic way to construct biological relationship as trees by comparing DNA/protein sequences.
Basic assumptions: sequences share a common ancestor, mutated from a common ancestor, mutations are rare.

Newick Format

```
((((A:0.21,B:0.21):0.28,C:0.49):0.13,D:0.62):0.38,E:1.00);
```

NEXUS Format

```
BEGIN TAXA;
        DIMENSIONS NTAX = 5;
        TAXLABELS
                Homo
                Pongo
                Macaca
                Ateles
                Galago ;
END;
BEGIN TREES;
        TRANSLATE
                1 Homo,
                2 Pongo,
                3 Macaca,
                4 Ateles,
                5 Galago ;
        TREE * UNTITLED = [&R]((((1:0.21,2:0.21):0.28,3:0.4
9):0.13,4:0.62):0.38,5:1);
END;
```

Possible Tree Topologies
k nodes has $(2k-3)!!$ rooted, $(2k-5)!!$ unrooted trees.

UPGMA Phylogenetic Tree Reconstruction
Unweighted Pair Group Method with Arithmetic Mean
1. Compute the distance between each pair of sequences
2. Treat each sequence as a cluster by itself
3. Merge the two closest clusters. The distance between two clusters is the average distance between all their sequences

$$d(C_i,C_j)=\frac{1}{|C_i||C_j|}\sum_{r\in C_i, s\in C_j} d(r,s)$$

4. Repeat 2 and 3 until only one cluster remains

Neighbor Joining

$$u(C_i)=\sum_j d(C_i,C_j), Q(i,j)=(r-2)d(C_i,C_j)-u(C_i)-u(C_j)$$

1. Start with each sequence as a cluster. All of them are connected to a hub, forming a star.
2. Find clusters i and j connected to the hub where $Q(i,j)$ is minimum among all cluster pairs
3. Insert a new internal node C_k, and connect it to C_i , C_j and the hub
Assign length $\frac{d(C_i,C_j)}{2}+\frac{u(C_i)-u(C_j)}{2(r-2)}$ to the edge C_iC_k (r is the number of clusters before the merge)
Assign length $\frac{d(C_i,C_j)}{2}+\frac{u(C_j)-u(C_i)}{2(r-2)}$ to the edge C_jC_k
For each node C_l, $d(C_k,C_l)=[d(C_i,C_l)+d(C_j,C_l)-d(C_i,C_j)]/2.$
4. Repeat 2 and 3 until all branch lengths are assigned, remove the hub and write down the distance.

Maximum Parsimony
Given a set of sequences and a rooted tree topology of the sequences, find the ancestral sequences, such that the total number of mutations along the branches is minimized.
For each internal node i with parent p and children l and r, determine the preference set S_i and final character C_i minimizing the total number of mutations.
1. For each leaf node i, set S_i to the character of the sequence
2. Upward phase: For each internal node i,
if $(S_l \cap S_r)=\{\}$ // l and r do not agree: take both sets
$S_i := S_l \cup S_r$
else // l and r agree on something: take the agreed part
$S_i := S_l \cap S_r$
3. Downward phase: First pick any C_{root} from S_{root}. Then for each

internal node i,
if $C_p \in S_i$ // p agrees with i on something: take it
$C_i := C_p$
else // p disagrees with i: use i's own preferences
$C_i :=$ choose one from S_i
Extended: majority vote, choose C_i from the characters that *exist in the largest number of sets among* $\{C_p\}$*,* S_l *and* S_r. Also, whenever there are multiple choices, we choose each in turn to *enumerate all optimal solutions*.

Maximum Likelihood
Given a tree topology, maximize the probability of producing the observed data by the model with parameters. Start with a random estimate of θ and apply "hill climbing" to increase the likelihood.

Motifs and Domains
Patterns that are overrepresented (unlikely to occur by chance), usually have known or predicted functional roles, and are evolutionarily conserved. A domain is assumed to possess certain functional or structural independence. A motif may not. A domain is usually larger than a motif.

DNA Binding Motifs
Promoters (around the transcription start site): To help the formation of the transcription machinery
Enhancers (usually further away from a gene): To enhance the expression of a gene
Silencers: To inhibit the expression of a gene
Insulators: To mark expression boundaries

Consensus Sequence
(i) Majority Vote (Problem: Information loss)
(ii) IUPAC nucleotide code

A: Adenine	K: G or T
C: Cytosine	M: A or C
G: Guanine	B: C or G or T
T (U): Thymine (Uracil)	D: A or G or T
R: A or G	H: A or C or T
Y: C or T	V: A or C or G
S: G or C	N: any base
W: A or T	. or -: gap

Regular expression: [AB] for any element, {n,m} for repeated count ([min, max])

Position Weight Matrix

	1	2	3	4	5
A	1/5	1/5	0/5	1/5	1/5
C	2/5	0/5	0/5	1/5	2/5
G	1/5	0/5	5/5	0/5	1/5
T	1/5	4/5	0/5	3/5	1/5

Pseudo-counts: add a small number to each count, to alleviate problems due to small sample size

Sequence Logo
Nucleotide with the highest probability on top
Total height of the nucleotides at the i-th position,

$$h_i = 2 + \sum_{x \in \{A,C,G,T\}} p_{i,x} \log_2 p_{i,x} - \frac{3}{2n \ln 2}$$

$p_{i,x}$: probability of character x at position i; n: number of sequences, height of nucleotide: $x = p_{i,x} h_i$.

Identifying over-represented motifs
1. Exhaustive search for all words of size up to k
2. Multiple sequence alignment
3. Integrating auxiliary information for finding active binding site: expression level, ChIP-chip, chromatin signals

Protein Domain Model: Pfam (HMM)
Four types of entries: family (collection of related proteins), domain (structural unit found in multiple protein contexts), repeat (stable only when multiple copies are present), motif: short unit outside globular domains.
Pfam A: high quality, manually curated
Pfam B: lower quality, automatically curated
Each family: an alignment of some representative *seed sequences* of the family, profile HMM (similar to PWM, also considers relationships among positions, constructed by HMMER3, used to scan all protein sequences in UniProtKB to find occurrences).
Additional information: domain architecture, phylogenetic tree of sequences, structural information

X-ome: A large amount of data related to X, or the whole set of X
X-omic: To study a large amount of data related to X
X-omics: The area of studying a large amount of data related to X
Workflow: 1. Production of data 2. Data processing: quality control, data normalization 3. Data analysis (pattern discovery) 4. Data annotation and comparisons: evaluation of statistical significance 5. Selection and summarization of results 6. Hypothesis formation 7. Experimental validation
Strengths of the omic approach: high-throughput: fast, less tedious, relatively inexpensive, comprehensive, relatively unbiased, easier to study interactions and combinatorial effects
Weaknesses: noise, secondary effects, lack of clear hypotheses, high initial cost for machines.

Genomic Sequencing
Sanger DNA sequencing: low throughput, high reliability, 300-1000 nt, commonly used in laboratory.
Second generation: parallel (platform - droplet/solid-phase, immobilization – primer/template/polymerase)
Third generation: longer reads, higher error/cost, single-cell sequencing
Cut the DNA into shorter fragments, randomly in overlaps, and perform de-novo assembly, or resequencing.
Raw data: single-end sequencing: one sequencing read (i.e., a short string) per fragment, or paired-end sequencing: two sequencing reads per fragment.
Quality score: how reliable each sequenced base is.

FASTQ File Format for raw reads
Sequence (FASTA) + Quality (Q)
Each sequence occupies four lines:

```
@SEQ_ID
GATTTGGGGTTCAAAGCAGTATCGATCAAATAGTAAATCCATTTGTTCAACTCACAGTTT
+
!''*((((***+))%%%++)(%%%%).1***-+*''))**55CCF>>>>>>CCCCCCC65
```

Phred score = (ASCII code of character) – 33;
If probability of base-calling error is p, then Phred score is $-\log_{10} p$.

de Bruijn Sequence Assembly
Considering k-mers of the subsequences: connect each two nodes in consecutive positions on a single sequence read.
Signals of errors: tips, bubbles, low-coverage paths.
Not very common to perform de novo assembly: high-throughput sequencing gives short reads, making de Bruijn graphs hard to simplify, databases are sufficient.
Mapping is more common by alignment to templates.
Efficient technique: suffix trie.

SAM File Format
Alignment:

```
Coor     12345678901234  5678901234567890123456789012345
ref      AGCATGTTAGATAA**GATAGCTGTGCTAGTAGGCAGTCAGCGCCAT
+r001/1        TTAGATAAAGGATA*CTG
+r002         aaaAGATAA*GGATA
+r003       gcctaAGCTAA
+r004                     ATAGCT..............TCAGC
-r003                            ttagctTAGGC
-r001/2                                        CAGCGCCAT
```

SAM format:

```
@HD VN:1.3 SO:coordinate
@SQ SN:ref LN:45
r001 163 ref  7 30 8M2I4M1D3M = 37  39 TTAGATAAAGGATACTG *
r002   0 ref  9 30 3S6M1P1I4M *  0   0 AAAAGATAAGGATA    *
r003   0 ref  9 30 5H6M       *  0   0 AGCTAA   *   NM:i:1
r004   0 ref 16 30 6M14N5M    *  0   0 ATAGCTTCAGC       *
r003  16 ref 29 30 6H5M       *  0   0 TAGGC    *   NM:i:0
r001  83 ref 37 30 9M         =  7 -39 CAGCGCCAT         *
```

M: alignment match; S: substitution (mismatch);
I: insertion; D: deletion.

Transcriptomics
RNA level does not perfectly correlate with protein level due to posttranscriptional regulation, RNA degradation, translation efficiency.
High-throughput methods: Microarrays, (i) design probes , (ii) convert RNA back to DNA, (iii) hybridization, (iv) fluorescent dye as measurable output;
For each gene, we design short sequences (25-75nt) unique to the gene. Binds to the complementary probe.
Microarray is data are relatively noisy: (i) cross-hybridization (binding to an unexpected probe); (ii) background signals, (iii) sensitivity to experimental condition.
cDNA sequencing (RNA-seq) (i) convert RNA back to cDNA (ii) sequence DNA, (iii) map back to genome, (iv) count number of reads from each gene. Better signal-to-noise ratio than microarrays, wider signal range, no prior knowledge of sequences.

Two-way Hierarchical Clustering
Euclidean distance (considers absolute expression levels):

$$d(x,y) = \sqrt{\sum_i (x_i - y_i)^2}$$

Pearson correlation (only trend matters):

$$r(x,y) = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_i (x_i - x)^2} \sqrt{\sum_i (y_i - y)^2}}$$

Nearest neighbor: kNN voting.

Genome Annotation
The process to catalog all functional elements in a genome and characterize their properties. Types of functional genomic elements: Protein coding genes (Transcripts, exons, introns, coding sequences (CDSs), untranslated regions (UTRs), ...), Non-coding RNAs.
Human genomic elements: NCBI RefSeq, Ensembl, UCSC, Gencode
Functional annotation: gene ontology, biological pathways
Three main sequence databases: DDBJ, EMBL, GenBank.
GeneCards, UCSC Genome Browser, UniProt
Gene Ontology (GO): an ontology for gene products produced by the Gene Ontology Consortium, 3 sub-ontologies: Molecular function (low-level functions of a gene product), Biological process (High-level processes that a gene product is involved), Cellular component (Which compartment can the gene product be found)
2 parts: the ontologies by directed acyclic graphs (DAG), some terms have multiple parents, edges indicate three types of relationships: is-a, part-of, regulates, and organism-specific instances (each gene can have 0, 1 or more annotated terms).
Evidence Codes: specifies how terms are used to annotate particular instances (Experimental, Computational analysis, Author statement, Automatically, Obsolete); Pathways: describe detailed mechanistic relationships between the objects.
KEGG: Kyoto Encyclopedia of Genes and Genomes Pathway Database: provides non-species-specific reference pathways, as well as species-specific versions; Different types of pathways: metabolic pathways, genetic information processing, environmental information processing, cellular processes, organismal systems, human diseases, drug development.

Functional Enrichment Analysis

	Genes in target set	Genes not in target set	Total
Genes annotated with a term	m_1	m_2	m_1+m_2
Genes not annotated with a term	n_1-m_1	n_2-m_2	$n_1+n_2-m_1-m_2$
Total	n_1	n_2	n_1+n_2

Null hypothesis: term and gene set are independent.

$$\mathbf{Pr}(m_1|H_0) = \frac{\binom{n_1}{m_1}\binom{n_2}{m_2}}{\binom{n_1+n_2}{m_1+m_2}}$$

Graph-Based Methods
Hierarchical relationship exists between different terms, use shortest path in the graph as distance, instead of common terms.

Multiple Hypothesis Testing
Multiply the p-value by the number of terms tested
Tools: AmiGO, BiNGO, DAVID (Database for Annotation, Visualization and Integrated Discovery) [ID conversion, Enrichment analysis, Multiple hypothesis testing correction], GSEA (Gene Set Enrichment Analysis) [Find the sets of genes annotated, Check the ranks based on phenotypic measure for calling gene sets, Compute statistic based on the ranks, Evaluate statistical significance by permuting phenotypic measure values].

Four Levels of Molecular Structures
Primary structure: the sequence
- Connections (chemical bonds) between molecules and atoms
- First-level constraints of the possible structures
- Orientation: DNA/RNA – 5' to 3'; Amino acids – Amino (N) terminus to carboxyl (C) terminus
- "Residue": the remaining amino acid after a water molecule is expelled
- Example: Strong covalent bonds hold together the phosphate-deoxyribose backbone of the DNA, and weaker hydrogen bonds hold together the two strands of the DNA double helix.

Secondary structure: local spatial conformation
DNA: Double helix

A-DNA	Right-handed	11bp per turn
B-DNA (common)	Right-handed	10.5bp per turn
Z-DNA	Left-handed	12bp per turn

RNA: Helices and single-strand loops, pseudoknots
Protein: α -helixes and β -sheets, coils/connectors

Tertiary structure: global, folds and domains
DNA: wrapped around nucleosomes formed by histone proteins, condensed form at the beginning of mitosis and meiosis
tRNA: 4 main arms - D loop, T loop, anticodon loop, acceptor arm attaching one amino acid
Protein: complex structures caused by weak forces (hydrogen bonds, and hydrophobic interactions), stronger forces (disulfide bonds between cysteines), classified by the CATH hierarchy.
Class: composition of secondary structures
Architecture: overall shape
Topology: connection of secondary structures
Homologous: with common ancestor

Quaternary structure: association of multiple molecules
Types: protein subunit-protein subunit, protein-protein, protein-DNA, protein-RNA, RNA-RNA
Function depends on structure: (i) as a material, (ii) binding (complementarity of interacting structures, formation of special bonds), (iii) functional group (e.g. catalytic site), (iv) determining localization (e.g. transporter membrane proteins)